

LIFEVISION

A pool camera that **notices**

Finds every person in the water and warns when
someone stays out of sight too long.

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THE PROBLEM

Drowning is **silent**

#1

cause of death for children
ages 1 to 4

2020

U.S. drowning rates began
rising again after decades of
decline

There is no splash and no scream. We chose drowning before falls on factory cameras because it is the most urgent case.



OUR PRODUCT

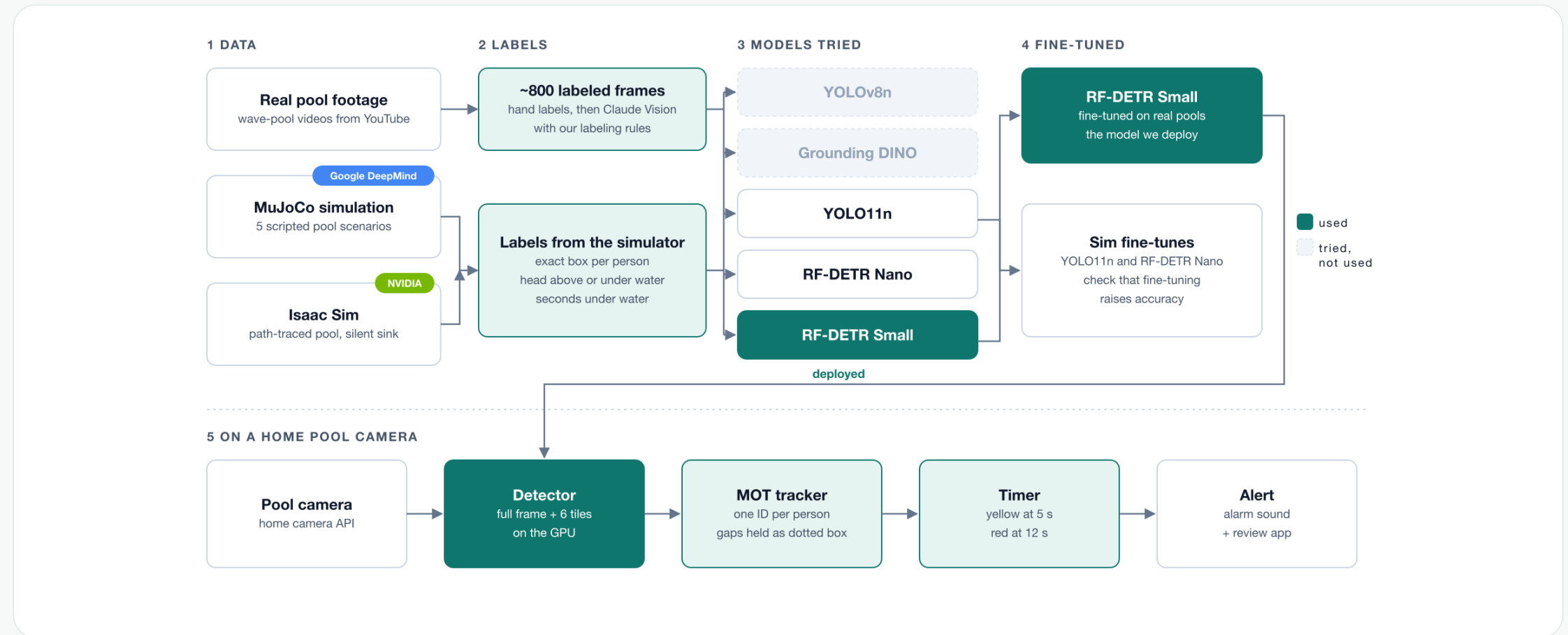
Uses the camera you already own

- ✓ Finds every person in the water
- ✓ Keeps one ID on each person
- ⚠ Out of sight 5 s: **yellow**. 12 s: **red**
- 👤 An adult still responds



HOW IT WORKS

The full pipeline



THE GAP

Bridging the gap between simulation and real pools

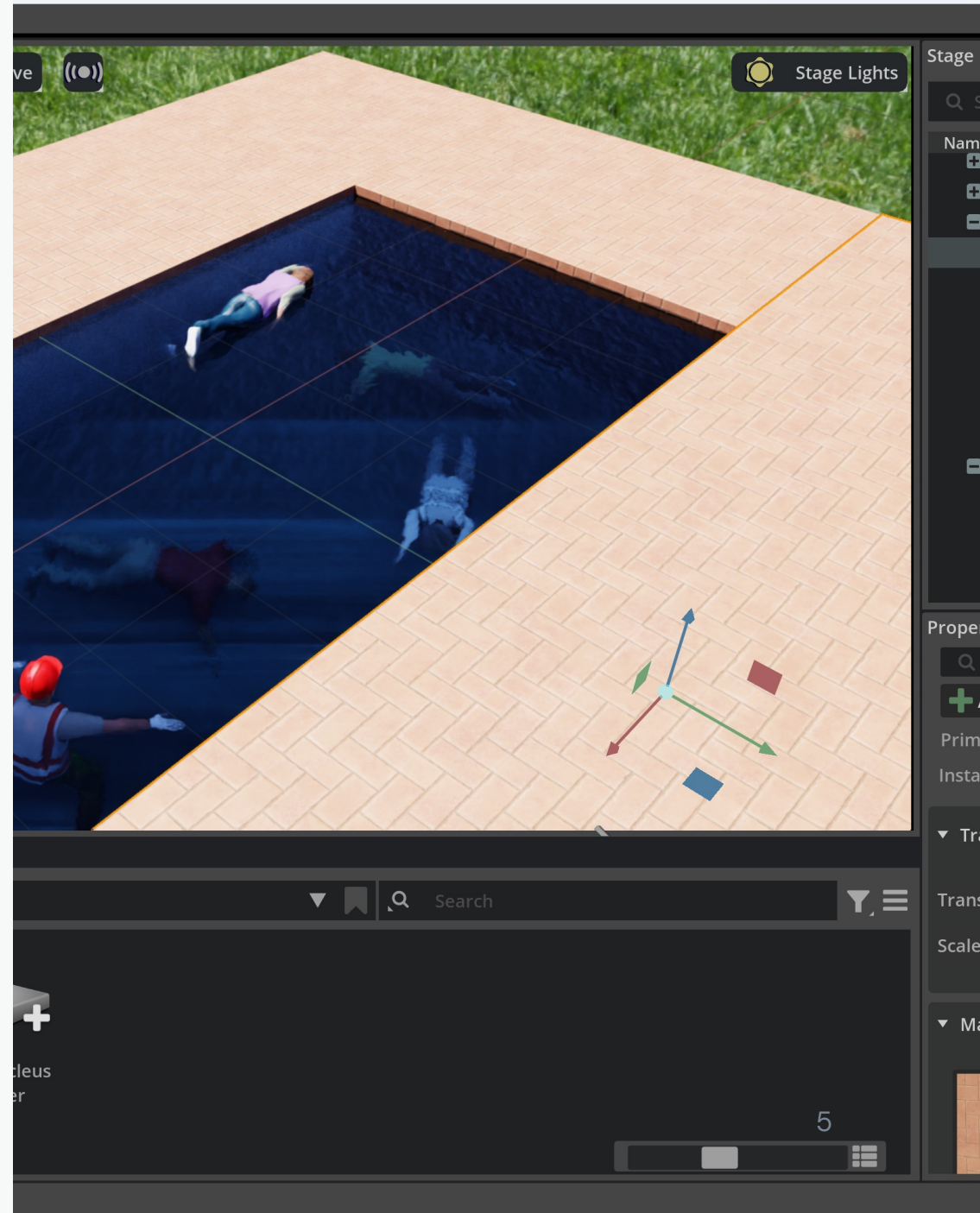
🔍 No dataset of children drowning exists

🔒 We can't film it

💡 So we simulated it, children and adults

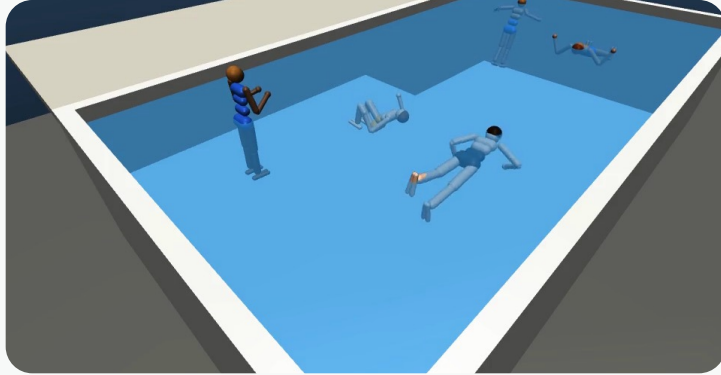
NVIDIA Isaac Sim

Google DeepMind MuJoCo



SIMULATION

Data we could not film



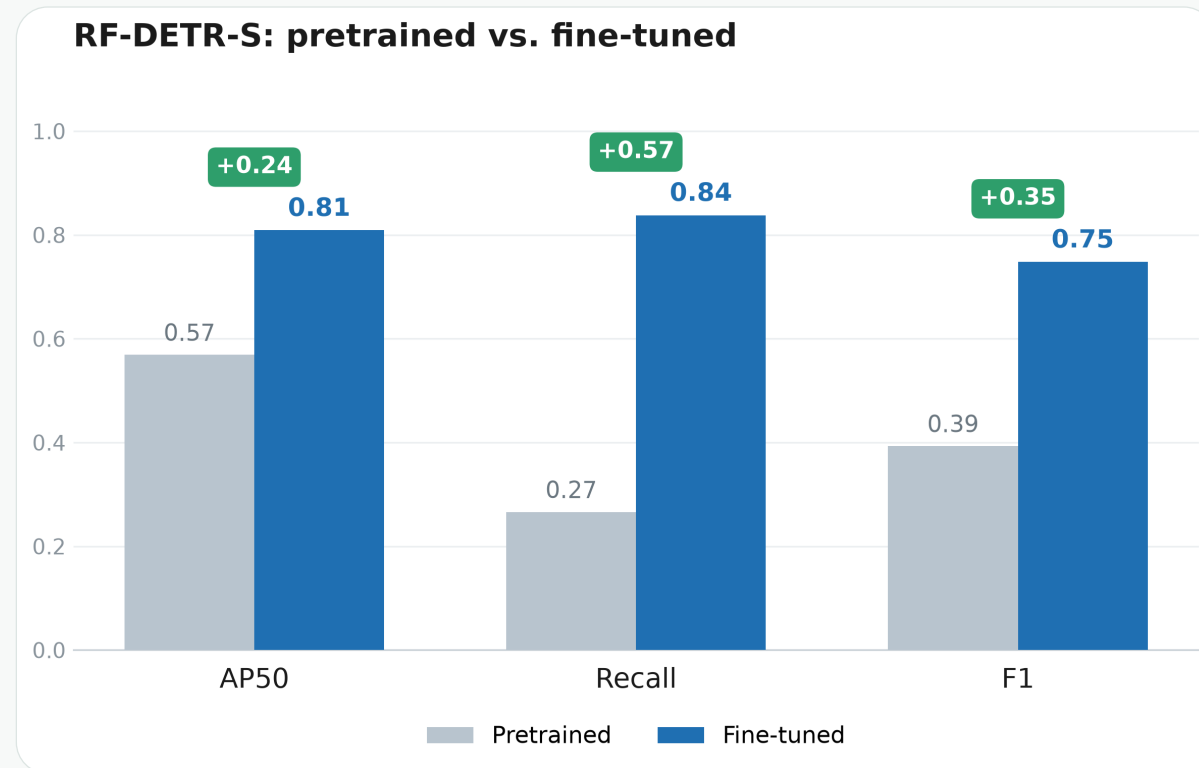
16% → 95%

people found with their head fully under water, stock vs fine-tuned YOLO11n (on simulation)

- ⚡ Much faster than finding footage at the right angle and resolution
- ✓ Labels come free: exact box and head depth

FINE-TUNING

Trained on a **real wave pool**



F1 score

0.75

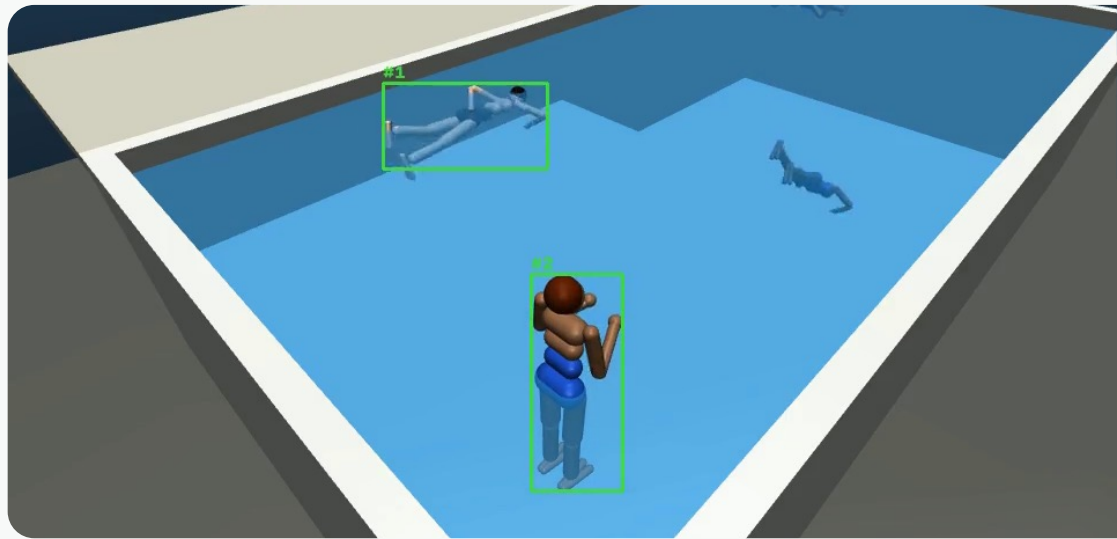
up from 0.39

- 👤 Hand-labeled pool frames
- ⚙️ Then Claude Vision with our rules
- ⚡ RF-DETR Small on a laptop GPU

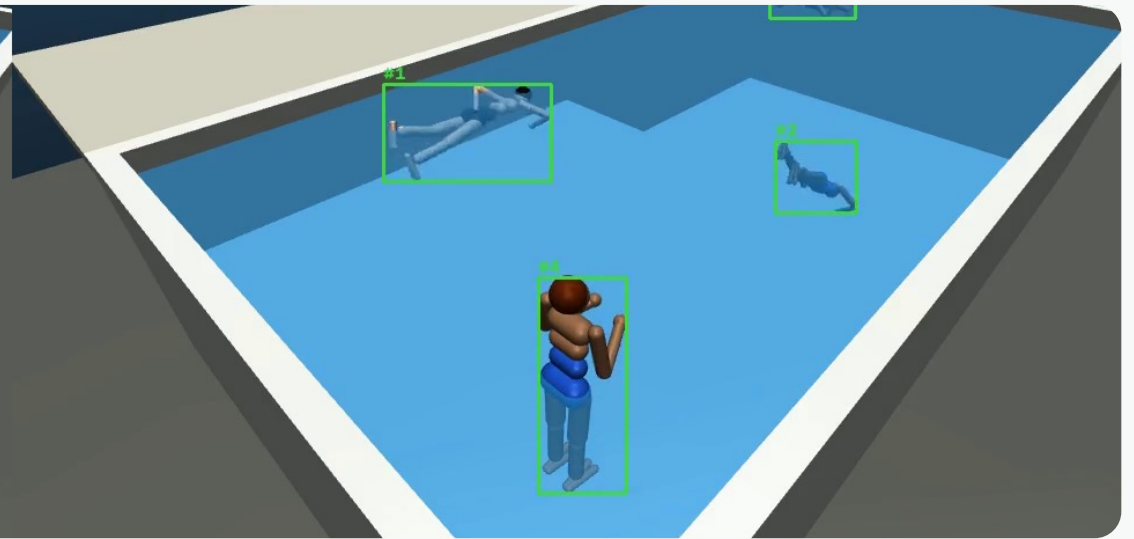
TRACKING

One ID per person

Stock model: 18 IDs for 4 people



Our tracker: 4 IDs for 4 people



Far-away swimmers blink in and out. A box missing for under a second is held as a **dotted box** and gets its ID back when it reappears. (on simulation)

LIVE DEMO

See it run

Detector, tracker, then timer, on a lifeguard rescue recording.



DEPLOYMENT

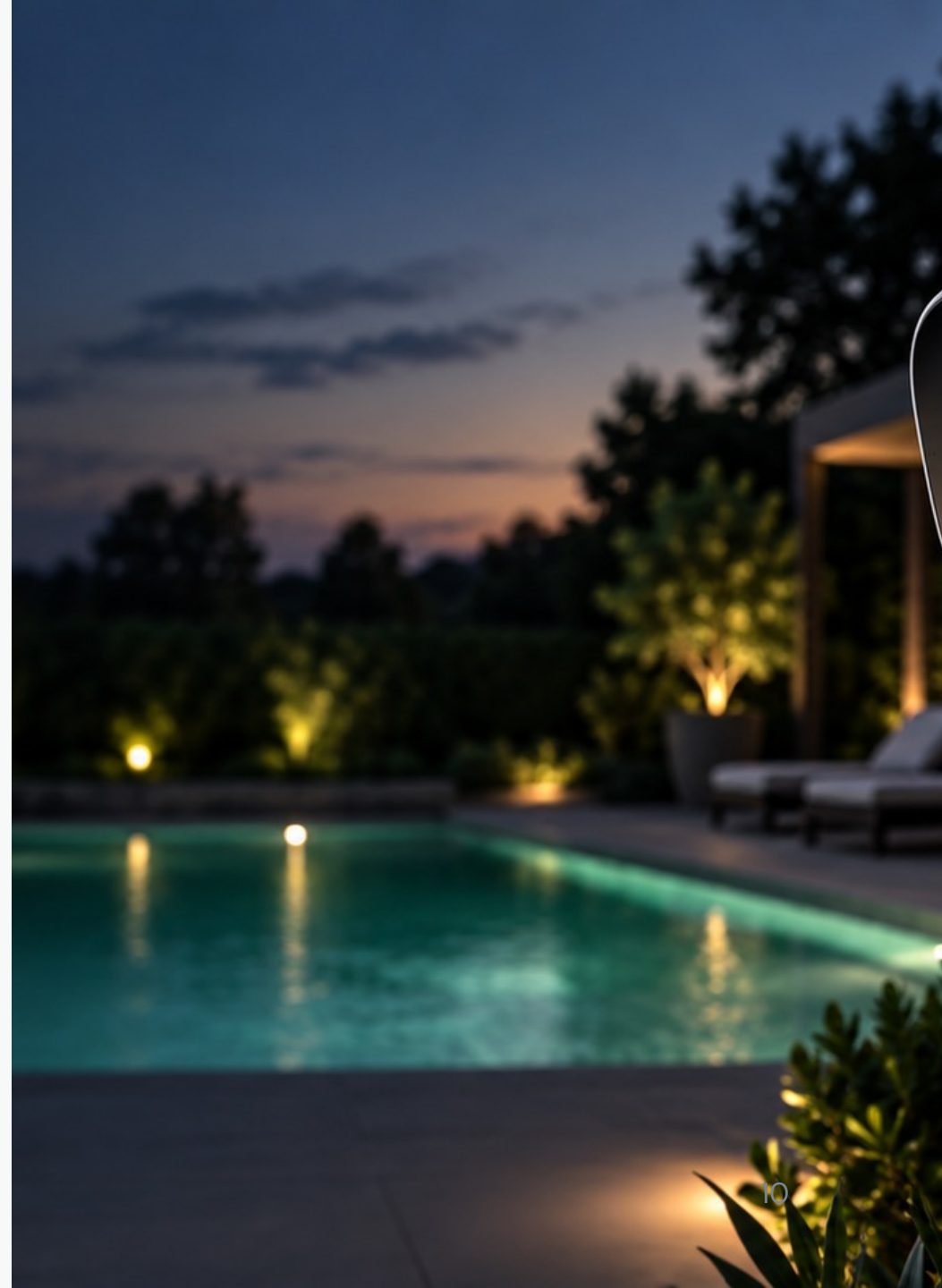
Two ways to **bring it home**

 **Runs locally.** Footage never leaves the house.

 **Our cloud.** Cheaper compute, faster processing.

 **Camera provider APIs** for live alerts with sound.

Alert timer set from lifeguard rescue guidelines. Shortened for the demo.



WHAT'S NEXT

From pools to **every camera**

- 📅 Keep fine-tuning for higher accuracy
- 🌐 More edge cases from Isaac Sim
- 🔒 More data through camera partners, privacy first

A supervision aid, not a replacement for watching children.

